

Data and Artificial Intelligence

Cyber Shujaa Program

Week 4 Assignment

Business Intelligence on Power BI

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Introduction

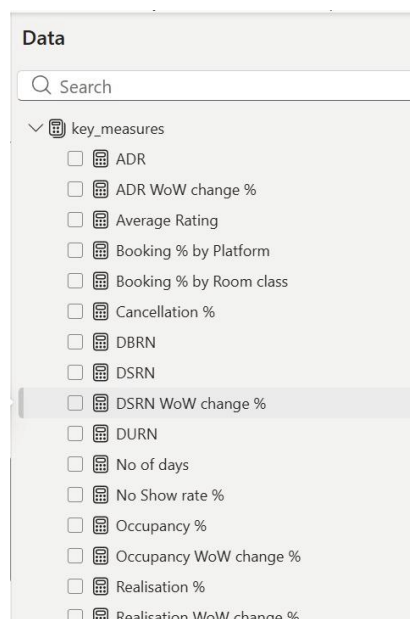
This week's assignment focused on developing hands-on experience in Business Intelligence using Microsoft Power BI, with a specific focus on the hotel management domain. The objective was to simulate a real-world business intelligence project by understanding hotel business operations, preparing and modeling data, performing analysis using DAX, and presenting insights through an interactive dashboard. The project was based on the end-to-end data analyst workflow provided by Codebasics and aimed at building a portfolio-ready Power BI solution that supports data-driven decision making in the hospitality industry.

The key goals of the assignment were to understand hotel business and client needs, load and transform multiple datasets, design a star-schema-based data model, create calculated columns and measures using DAX, visualize insights using an interactive dashboard, and publish the final report for sharing.

Tasks Completed

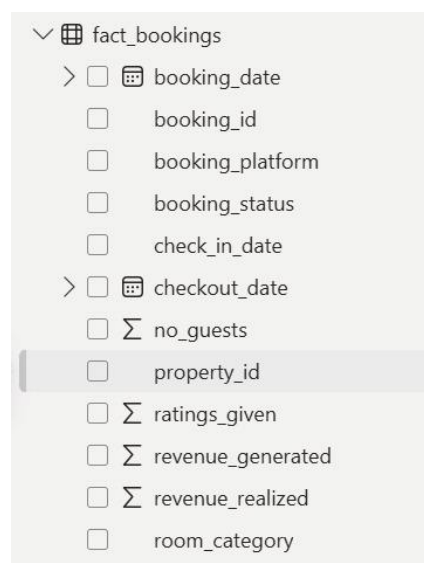
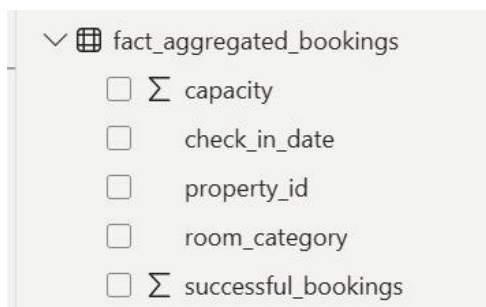
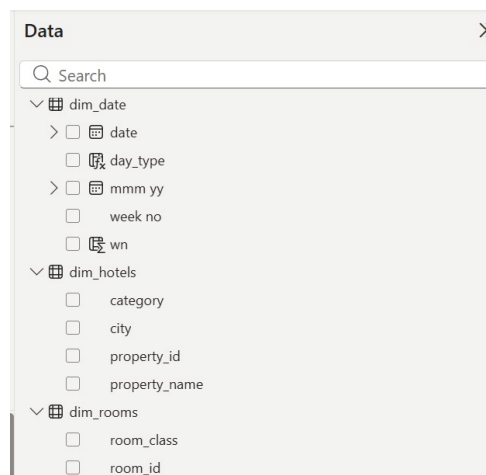
Step 1: Understanding the Business Requirements

The first step involved understanding the hotel business context and identifying key performance indicators relevant to hotel management. These included metrics such as occupancy rate, revenue, average daily rate (ADR), revenue per available room (RevPAR), booking trends over time, and room performance. This understanding guided the selection of transformations, measures, and visualizations needed to support managerial decision-making.



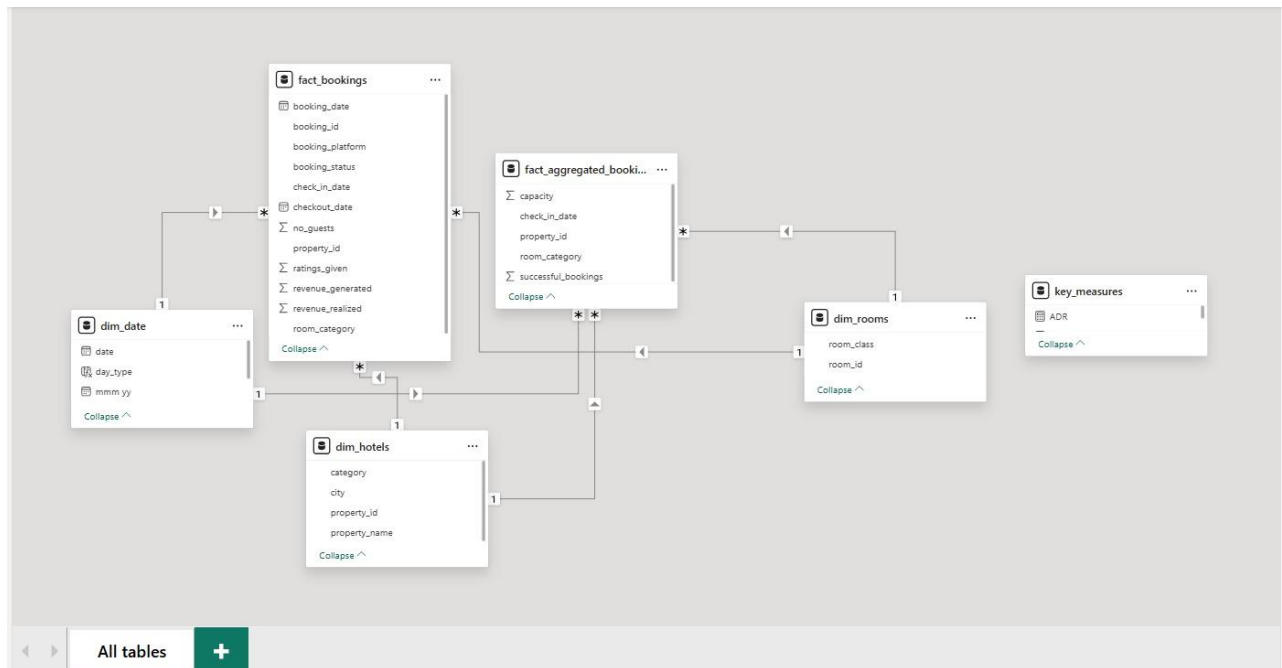
Step 2: Data Loading and Transformation

Multiple datasets were loaded into Power BI, including fact and dimension tables such as the bookings data, dim_date, dim_rooms, and other supporting tables. Power Query was used extensively to clean and transform the data. This included removing duplicates, handling missing values, correcting data types, renaming columns for clarity, and creating new derived columns where necessary. Each transformation step was documented clearly to ensure transparency and reproducibility.



Step 3: Data Modelling

A star schema was implemented to structure the data model effectively. The central fact table containing booking and revenue data was connected to relevant dimension tables such as date and room dimensions using appropriate one-to-many relationships. Proper relationships were established to ensure accurate filtering and aggregation across visuals.



Step 4: DAX Calculations

Data Analysis Expressions (DAX) were used to create calculated columns and measures essential for analysis. Measures such as Total Bookings, Occupancy %, Average Daily Rate (ADR), and RevPAR were created to support performance tracking. Meaningful naming conventions were used, and calculations were structured for readability and maintainability.

```

1 ADR = DIVIDE([Revenue], [Total Bookings], 0)

1 RevPAR = DIVIDE([Revenue], [Total Capacity])

1 Occupancy WoW change % =
2 Var selv = IF(HASONEFILTER(dim_date[wn]), SELECTEDVALUE(dim_date[wn]), MAX(dim_date[wn]))
3 var revcw = CALCULATE([Occupancy %], dim_date[wn] = selv)
4 var revpw = CALCULATE([Occupancy %], FILTER(ALL(dim_date), dim_date[wn] = selv - 1))
5
6 return
7
8
9 DIVIDE(revcw, revpw, 0) - 1

```


Link to Project:

https://drive.google.com/file/d/1wBXanLPodX6ymox7g9ipvlqeFpAUYe7T/view?usp=drive_link

Conclusion

This assignment provided valuable practical experience in applying Business Intelligence concepts using Power BI within a real-world hotel management context. Through data preparation, modelling, DAX analysis, and dashboard visualization, I gained a deeper understanding of how data-driven insights can support strategic and operational decision making. I look forward to applying these skills in more advanced BI and data analytics projects in future coursework and professional settings.